



Problem Solving2

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Q1 Harshad number

10 marks

A number is called a Harshad number (or Niven number) if it is divisible by the sum of its digits. Example: $18 \rightarrow \text{digit sum} = 9, 18 \% 9 == 0$. Write a Python program to read a number N from the user and check whether it is a Harshad number. Also print all Harshad numbers between 1 and 500.

Your Answer

```
# Check whether a number is a Harshad number

n = int(input("Enter a number: "))

temp = n
digit_sum = 0
```



Q2 GCD

10 marks

The following Python program is intended to find the GCD (Greatest Common Divisor) of two numbers using the Euclidean algorithm. It contains FOUR errors. Identify each error and write the corrected line.

```
a = int(input("Enter first: ")) b = int(input("Enter second: "))
while b = 0: temp = a a = b b = temp % a
print("GCD is", a)
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 a = int(input("Enter first: "))
2 b = int(input("Enter second: "))
3
4 while b != 0:
5     temp = a
6     a = b
7     b = temp % b
8
9 print("GCD is", a)
```

Test Input

24
36

Output

```
Enter first: Enter second: GCD is
12
```



```
all seven blanks.  
def find_pairs(lst, target):  
    seen = _____ pairs = []  
    for num in _____:  
        complement = target - _____  
        if complement in _____:  
            pairs.append((complement, num))  
            _____add(num)  
    return _____  
data = [int(x) for x in input("Enter numbers: ").split()]  
t = int(input("Target: "))  
result = _____(data, t)  
print("Pairs:", result)
```

Your Answer

```
def find_pairs(lst, target):  
    seen = set()  
    pairs = []  
  
    for num in lst:  
        complement = target - num
```

 Save Answer



Q4 Bitwise Operators

10 marks

Trace the following Python program step by step. Write the exact output. Also state the data type of each variable after assignment.

```
a = 0b1101
b = 0o25
c = 0x1A
result = (a & b) | c
print(f"a={a}, b={b}, c={c}, result={result}")
print(bin(result), oct(result), hex(result))
print(result >> 2, result << 1, ~result)
```

Your Answer

```
a = 0b1101
b = 0o25
c = 0x1A

result = (a & b) | c
```

 Save Answer



Q5 binary search

10 marks

The following Python program tries to perform binary search on a rotated sorted array. It has FOUR errors. Identify each error with the line number and rewrite the correct program.

```
def rotated_binary_search(arr, target):  
    low = 0  
    high = len(arr) - 1  
    while low <= high:  
        mid = (low + high) // 2  
        if arr[mid] == target:  
            return mid  
        if arr[low] <= arr[mid]:  
            if target >= arr[low] and target < arr[mid]:  
                high = mid - 1  
            else:  
                low = mid + 1  
        else:  
            if target > arr[mid] and target <= arr[high]:  
                low = mid + 1  
            else: high = mid - 1  
    return -1  
data = [4,5,6,7,0,1,2]  
t = input("Target: ")  
result = rotated_binary_search(data, t)
```




Your Fixed Code

```
1 def rotated_binary_search(arr, target):
2     low = 0
3     high = len(arr) - 1
4
5     while low <= high:
6         mid = (low + high) // 2
7
8         if arr[mid] == target:
9             return mid
10
11        if arr[low] <= arr[mid]:
12            if target >= arr[low] and target < arr[mid]:
13                high = mid - 1
14            else:
15                low = mid + 1
16        else:
17            if target > arr[mid] and target <= arr[high]:
18                low = mid + 1
19            else:
20                high = mid - 1
21
22    return -1
23
24 data = [4, 5, 6, 7, 0, 1, 2]
25 t = int(input("Target: "))
26 result = rotated_binary_search(data, t)
27 print("Found at:", result)
```

Test Input

0
4

Output

